In this report, we will consider the cost-effective control of soil transmitted helminths (STH, intestinal worms). A deterministic model is used to describe the transmission of STH between children and adults via an infectious source. The ODEs of the model and assumptions of distributions for worm distributions in people and hosts are given in the original document.

Endemic Equilibrium: We first compute ℓ in terms of the other model parameters at equilibrium. For equilibrium, we must have $\dfrac{dM_c}{dt} = \beta_c \ell - \sigma M_c = 0$ and $\dfrac{dM_a}{dt} = \beta_a \ell - \sigma M_a = 0$ i.e. $\overline{M_a} = \dfrac{\beta_a \ell}{\sigma}$ and $\overline{M_c} = \dfrac{\beta_c \ell}{\sigma}$, where $\overline{M_a}$, $\overline{M_c}$ are equilibrium values. Additionally, we must also have $\dfrac{d\ell}{dt} = 0$, therefore by substituting our equilibrium values of M_c and M_a into the equation formed by $\dfrac{d\ell}{dt} = 0$, we yield an equation in terms of ℓ, specifically:

$$\mu \ell = \frac{R_0 \sigma \mu}{(\beta_c \rho n_c + \beta_a (1-\rho) n_a)} \cdot \left[\phi(\overline{M_c} f(\overline{M_c}; k, z) n_c \rho + \phi(\overline{M_a}) f(\overline{M_a}; k, z) n_a (1-\rho) \right]. \quad (1)$$

We make an assumption that $\phi \approx 1$ for simplification and then use the `uniroot` to determine the solution for each posterior parameter set containing tuples of (R_0, k) values. We obtain the following endemic equilibrium distributions:

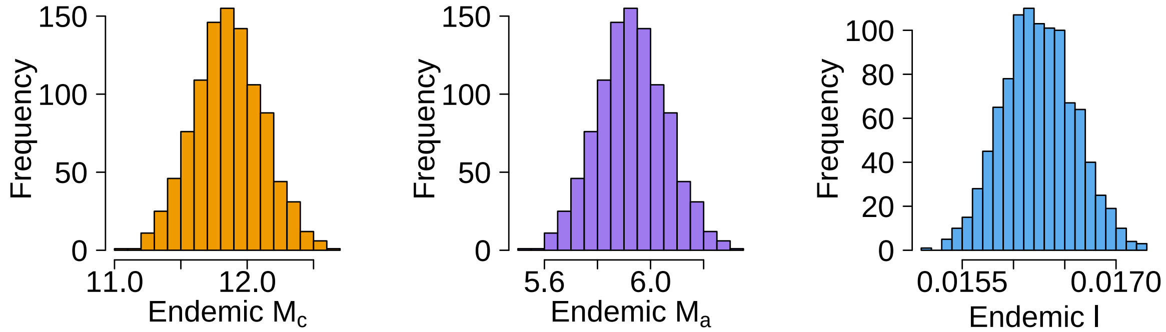

Figure 1: Endemic equilibrium distribution for the mean worm burden in children (M_c), and adults (M_a) and the infectious source (ℓ) after using `uniroot` and assumption $\phi \approx 1$ to solve for ℓ, using 1000 posterior parameters

The mean values of the distribution of endemic equilibrium are given by the following table:

M_c	M_a	ℓ
11.8578	5.9289	0.0162

Table 1: Mean values for endemic equilibrium of M_c, M_a and ℓ using 1000 posterior parameters.

After doing so, we use our numerical approximation as an initial condition i.e. $\left(\overline{M_c}, \overline{M_a}, \ell\right) = \left(\dfrac{\beta_c \ell}{\sigma}, \dfrac{\beta_a \ell}{\sigma}, \ell\right)$ to run our complete model to obtain a solution closer to the true endemic equilibrium. We run the simulations for each posterior parameter set for 20 years to ensure convergence. This took 16 minutes and 21 seconds to run. Upon doing so, we obtain the following endemic equilibrium distributions:

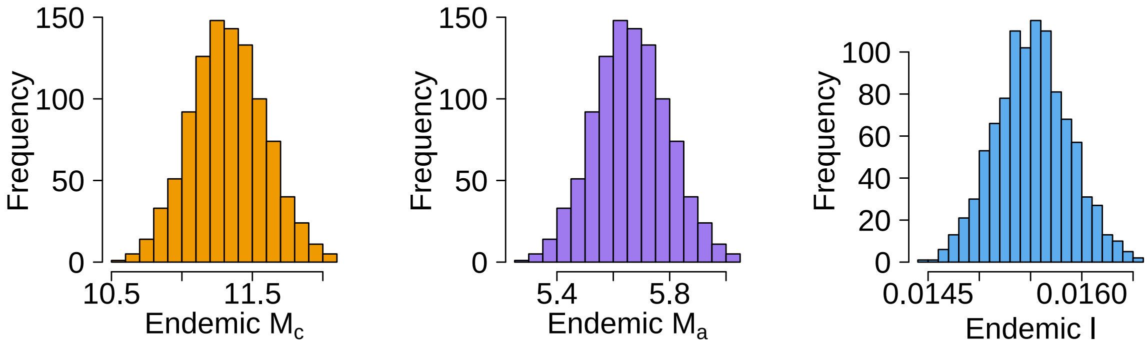

Figure 2: Endemic equilibrium distribution for the mean worm burden in children (M_c), and adults (M_a) and the infectious source (ℓ) after using `uniroot` and assumption $\phi \approx 1$ to solve for ℓ.

The mean values of the distribution of endemic equilibrium, after using the model, are given by the following table:

M_c	M_a	ℓ
11.348	5.674	0.0155

Table 2: Mean values for endemic equilibrium, after using the model for better accuracy, of M_c, M_a and ℓ using 1000 posterior parameters.

CHEERS Checklist: *Title*: A CEA with a deterministic model and gamma-distributed costs compares no intervention, intervention 1 (annual MDA for children), intervention 2 (biannual MDA for children), and intervention 3 (annual MDA for children and adults). Interventions are feasible for a low resource setting hence have been chosen.

Context: The study evaluates the cost-effective control of Ascaris lumbricoides (roundworm) infections using a deterministic transmission model for children and adults.

Study Context: Determine the most cost-effective approach for reducing worm burden and disease impact.

Practical Relevance: The findings guide policy decisions by assessing the health benefits, costs, and feasibility of different intervention strategies for optimal resource allocation.

Study Population: Demographically, 100,000 people 30% children, 70% adults. 85% school attendance, lower adult participation 20% in MDA application. Worm burden varies with high intensity (≥ 30 worms) causing significant disability weight (0.15), moderate ($15 - 29$ worms) has 0.01 weight and low intensity (< 15 worms) negligible.

Setting and Location: Endemic, low-resource region where STH are widespread and no prior interventions have been implemented. Since only 20% adults get MDA, suggests the location is limited by healthcare resources.

Perspective: The study adopts a healthcare provider and societal perspective to evaluate cost-effectiveness. This approach considers direct medical costs for policymakers and broader economic impacts like DALYs and productivity loss, ensuring a comprehensive analysis for decision-making in resource-limited settings.

Time Horizon: 20 years with 3% discounting, appropriate because it captures the long term effects of interventions. We are assessing sustained impact. Discounting required due to long time horizon so monetary value will change drastically.

Selection of outcomes: The study measures benefits through worm burden reduction and DALYs averted. Harms include intervention costs. These were measured as part of our CEA by running our model on 1000 posterior parameter simulations.

Currency, price date, conversion: We use standard US dollars currency and a unit cost of MDA dose is \$1.50 modelled using $\Gamma(4.5, 1/3)$. Additional scaling to account for difficulties of community administration in adults is modelled by $\Gamma(45, 2/30)$. Our plausible strategy of providing filtered water has cost per year of \$5.00 modelled with uncertainty as $\Gamma(66, 5/66)$.

Description of model: This model uses a set of deterministic ODEs to model the transmission of STH into children and adults separately. We model costs with uncertainty and use a set of posterior parameters for k and R_0 to create uncertainty within our ODEs. The model uses 3% discounting to value price across a 20 year time horizon.

Characterising Heterogeneity: We define two sub-groups within our population: children and adults. We account for the difficult of administrating MDA into adults as oppose to children who are available through schools by using weights of $g = 0.85$ and $g = 0.20$ respectively.

Characterising Distributional Effects: Additionally, we separate infectious cases by the number of worms into low, moderate and high intensity cases by creating subgroups based on < 15 worms, $15 - 29$ worms and ≥ 30 worms.

Effect of uncertainty: Our dose of MDA may come from multiple sources which have varying prices in the market and to account for this uncertainty we use a distribution as previously described from which we draw our prices in our model. Although this affects a single iteration, our cost-effective analysis is done over 1000 realisations so generally the output results will approximately use a mean value. The uncertainty allows us to produce CEAC curves to give probabalistic interpretations of cost-effectiveness.

Q1(i):

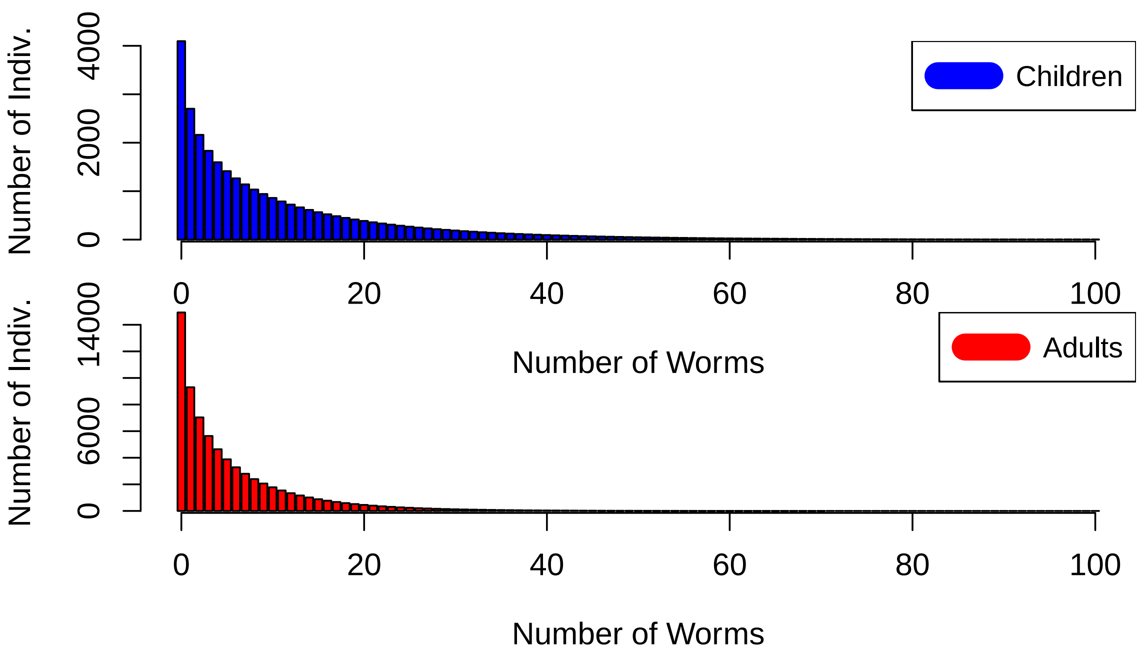

Figure 3: Comparison of pre-control worm burden distributions in children (left) and another related distribution (right). The figures use mean M_c, M_a, and k values from our posterior distribution.

The number of individuals in the "moderate" and "high" intensity infection categories is given in the following table:

	Moderate	**High**
Children	5,887	3,062
Adults	7,153	1,211

Table 3: Number of individuals (children and adults) in the "moderate" and "high" intensity infection categories.

Q1(ii):

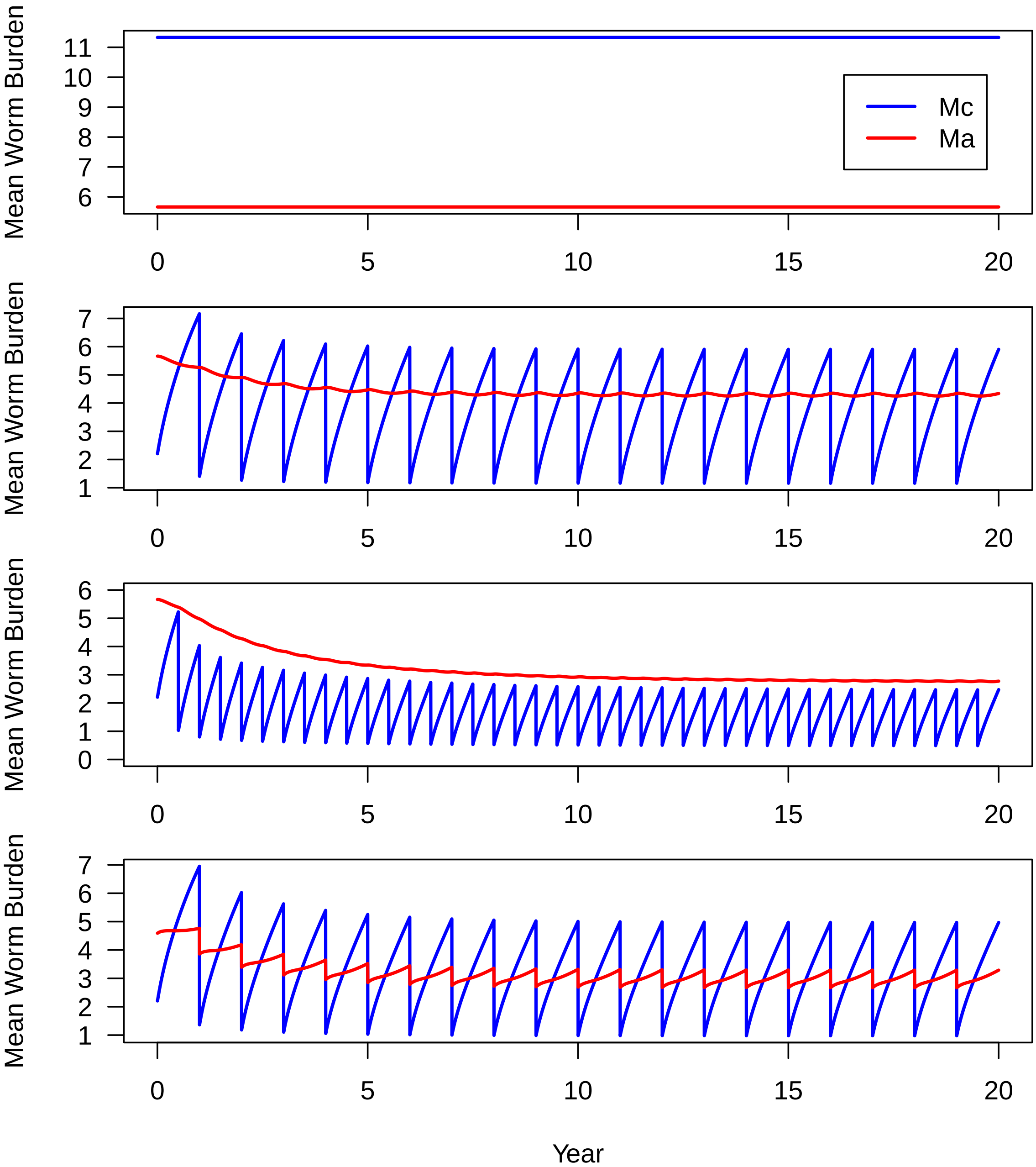

Figure 4: Comparator strategy; Annual MDA for children strategy; Biannual MDA for children; Annual MDA for children and adults. The mean worm burdens for children (blue) and adults (red) are plotted as the infective dynamics.

The comparator strategy involves doing nothing as a point of reference for the magnitude of the transmission of STH. Annual MDA (Intervention 1) for children involves giving the children class MDA instantaneously at the start of each year. Biannual MDA (Intervention 2) for children involves giving children MDA instantaneously at the start and mid-point of each year. Here we compare if doubling the doses improves cost-effectiveness or if effectiveness becomes a limiting factor. Finally, we apply MDA in both children and adults annualy (Intervention 3) to see if the additional costs of giving MDA to adults yields a better cost-effectiveness to our previous two strategies.

Q1(iii): We aim to compute the undiscounted mean DALYs per year for all the strategies outlined above. For any given strategy, we start by calculating the average values of M_c and M_a for each year. This is done by summing their respective values between years i and $i+1$ and dividing by 365. This gives the person years metric.

Next, we use these mean values to determine the proportion of people with exactly x worms. For each year, we iterate over the range $i \in [15, 29]$, computing the proportion of individuals with exactly i worms. We then multiply this by $N \cdot n_c$ to estimate the expected number of people with that worm count and apply the appropriate disability weighting. Since infections in the range $[15, 29]$ are classified as "moderate-intensity," we use a disability weight of 0.01. This process is performed separately for both children (M_c) and adults (M_a).

We then repeat the procedure for high-intensity infections, iterating over the range $i \in [30, 100]$ and applying a disability weight of 0.15. Summing the expected values from both moderate- and high-intensity infections gives the total DALYs for that year. By repeating this process across all years, we obtain the final mean DALYs per year for that strategy. This method is applied across all strategies. Observe we avoid deaths by mortality as assumed in the worksheet so our YLL = 0.

After applying this across all `1000` posterior parameter sets from our endemic equilibrium computations before, we calculate the mean DALYs alongside a confidence interval of `95%`. The results are shown in the following graph:

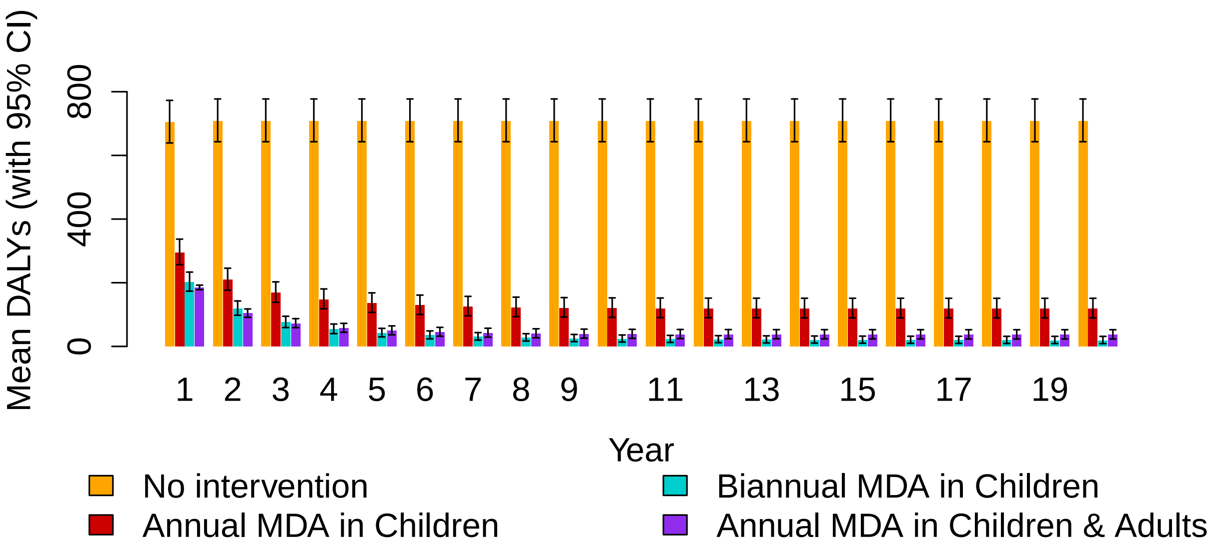

Figure 5: Mean DALYs per year with a 95% confidence interval for each strategy.

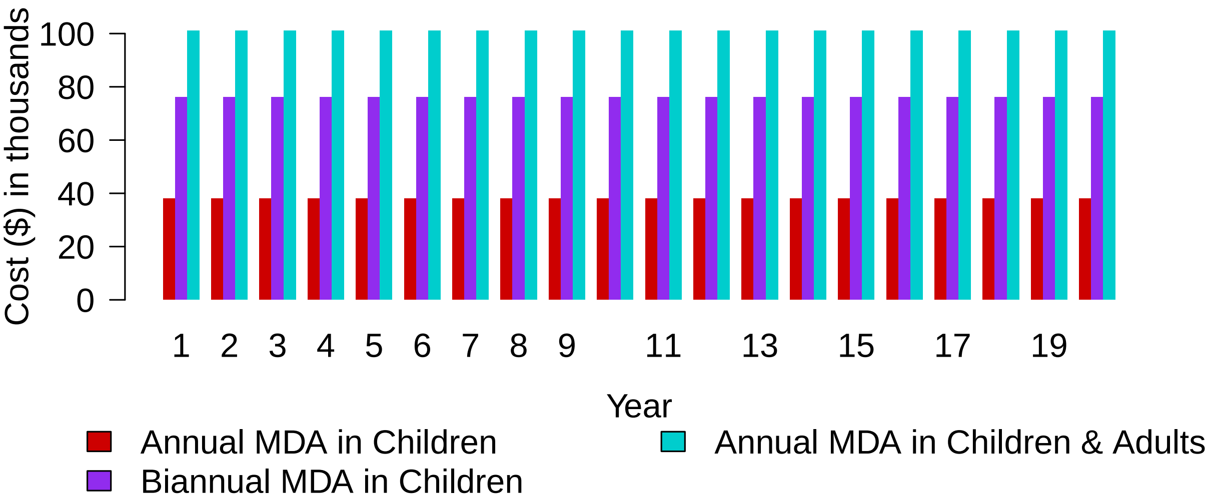

Figure 6: Undiscounted costs per year for each strategy.

Q1(iv): Table of discounted Δ costs, mean Δ DALYs and corresponding ICERs:

Intervention	Cost	DALYs	Δ Costs	Δ DALYs	ICERs
Comparator	0	10,844	0	0	Referent
Intervention 1	582,996	2,191	582,996	8,653	67.4
Intervention 2	1,165,992	721	1,165,992	10,123	396.6
Intervention 3	1,549,260	855	1,549,260	9,989	Dominated

Table 4: Cost-effectiveness analysis results. A table of costs, DALYs, discounted costs (Δ Costs), mean discounted DALYs (Mean Δ DALYs) and corresponding ICERs. Intervention 1 corresponds to annual MDA in children, intervention 2 corresponds to biannual MDA in children and intervention 3 is annual MDA in children & adults.

Q1(v): The cost-effectiveness plane for all three interventions:

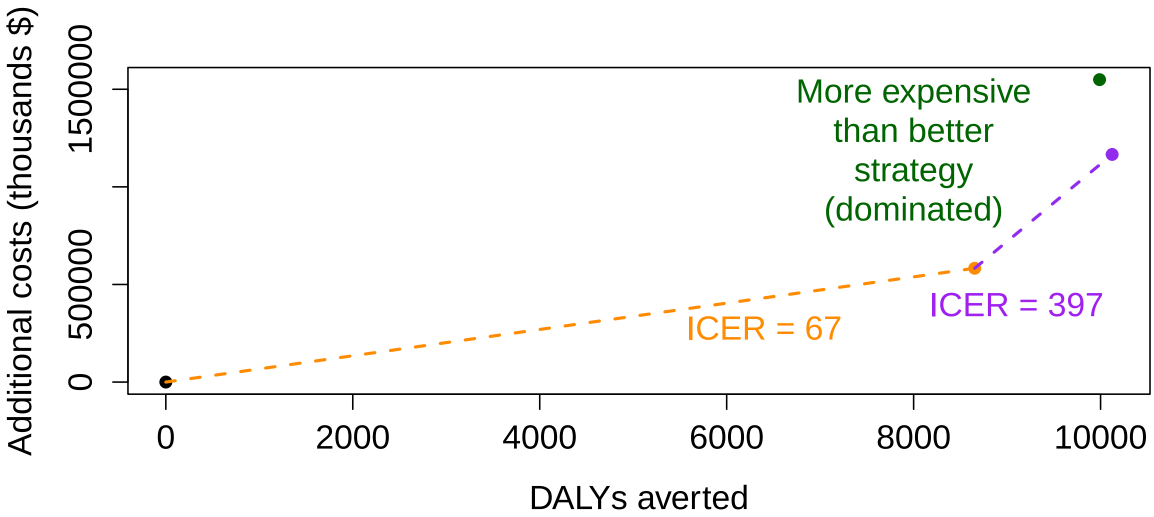

Figure 7: Cost-effectiveness plane for our three strategies.

Q1(vi): The plot of cost-effectiveness acceptability curves and frontier for informing our recommendation to the policy maker:

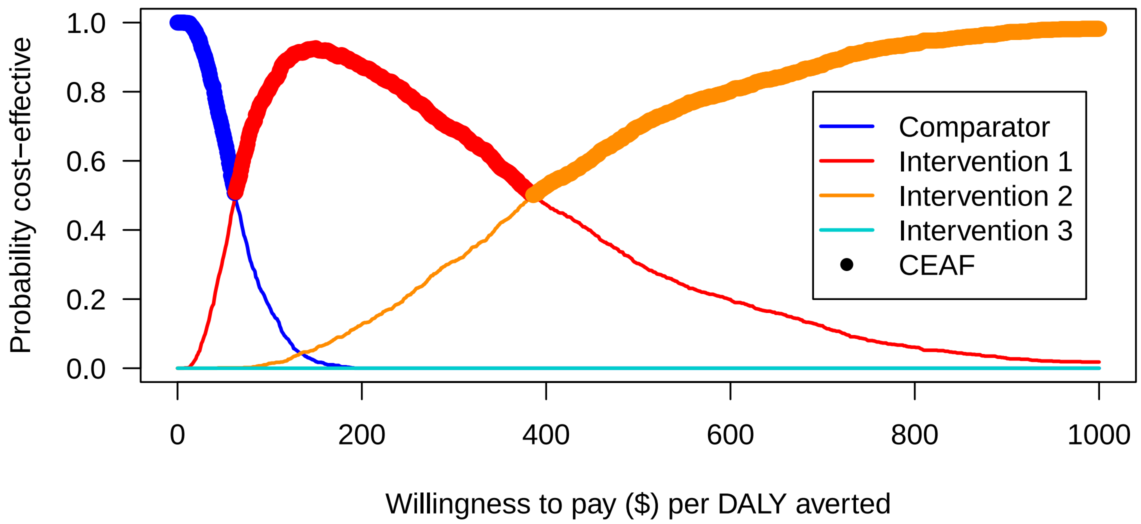

Figure 8: Cost-effectiveness acceptability curves for each strategy with intervention 1 being annual MDA in children, intervention 2 being biannual MDA in children and intervention 3 being annual MDA in children and adults.

Q1(vii): *Recommendation to a policy maker*: From our CEAC curves and the CEAF, if our WTP is $\leq \$62$ then apply no intervention. If our WTP is $> \$62$ and $\leq \$385$ then apply intervention 1 which is annual MDA in children. If our WTP is $> \$385$ then apply intervention 2 which is biannual MDA in children.

Q2a: Another plausible intervention is cleaning the infectious source—contaminated water and soil containing STH worms. To reflect this in our model, we apply a 95% reduction to the infectious source term. The first strategy applies this reduction annually, representing the distribution of treated water to families each year. The second strategy provides filtered water biannually. We assume each child and adult belongs to a separate household.

Q2b: A justified cost is \$5 as sourced from https://gh.bmj.com/content/6/12/e007361, this is the annual cost per household for water. We model this with uncertainty as a gamma distribution $\Gamma(66, 5/66)$. *Q2c*: We now perform our CEA using all five strategies and report a CEA table as well as a CEAC curve to make an updated recommendation.

Intervention	Cost	DALYs	Δ Costs	Δ DALYs	ICERs
Comparator	0	10,844	0	0	Referent
Intervention 1	582,996	2,191	582,996	8,653	67.4
Intervention 2	1,165,992	721	1,165,992	10,123	396.6
Intervention 3	1,549,260	855	1,549,260	9,989	Dominated
Intervention 4	7,668,838	5,550	7,668,838	5,295	Dominated
Intervention 5	15,337,676	2,253	15,337,676	8,591	Dominated

Table 5: Cost-effectiveness analysis results. Intervention 1, 2, 3 as before. Intervention 4 corresponds providing filtered water annually. Intervention 5 corresponds to providing filtered water biannually.

From the results above, since both of our additional interventions were dominated by intervention 2, the CE plane will not be different. Furthermore, our new recommendation remains the same based on the following CEAC curves for all 5 strategies:

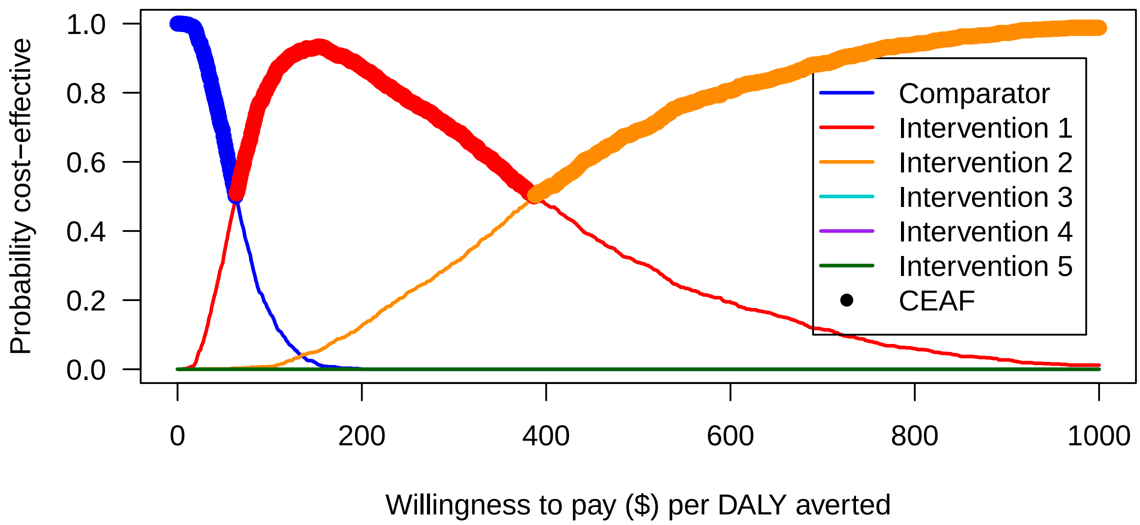

Figure 9: Cost-effectiveness acceptability curves for the three original strategies and two newly proposed strategies based on a water filtering intervention.

The NMB is too low due to the large ΔCosts for the newly proposed strategies. This makes the NMB much more negative compared to the other strategies and thus rejected in our probability for cost-effectiveness computation. Since the shape of the CEACs remain the same, based on ICER analysis, our recommendations also remain the same.

Q2d: Our CEA primarily focusses on the cost-effectiveness of implementing each strategy, however, there could be other factors influencing a policy makers decision. One that is captured within our model is:

- **Health equity and disparities**: In our model, we account for the disparity in administrating MDA in children compared to adults using the weights `g = 0.85 (children) and g = 0.2 (adults)`. Therefore, a policy maker may make a decision based on feasibility of giving the drug and this parameter can be altered to reflect this change.

Two factors that are not captured by our model but can influence a policy makers decision are:

- **Political feasibility and public opinion**: If an intervention is politically unpopular or against public interest (as often is the case with giving drugs) then it may not be considered by a policy maker. This is not captured in our model as we assume children and adults are willing to take the drug provided they have the access.

- **Legal and ethical considerations**: Our model is flexible whereby we can try strategies where we administer a drug on a monthly interval, and certainly this would improve the results, however, there could be ethical and legal considerations about giving such a large quantity of drugs to children and adults which is not captured by our model. There could be side effects which we don't account for or legal requirements about dosage amounts.